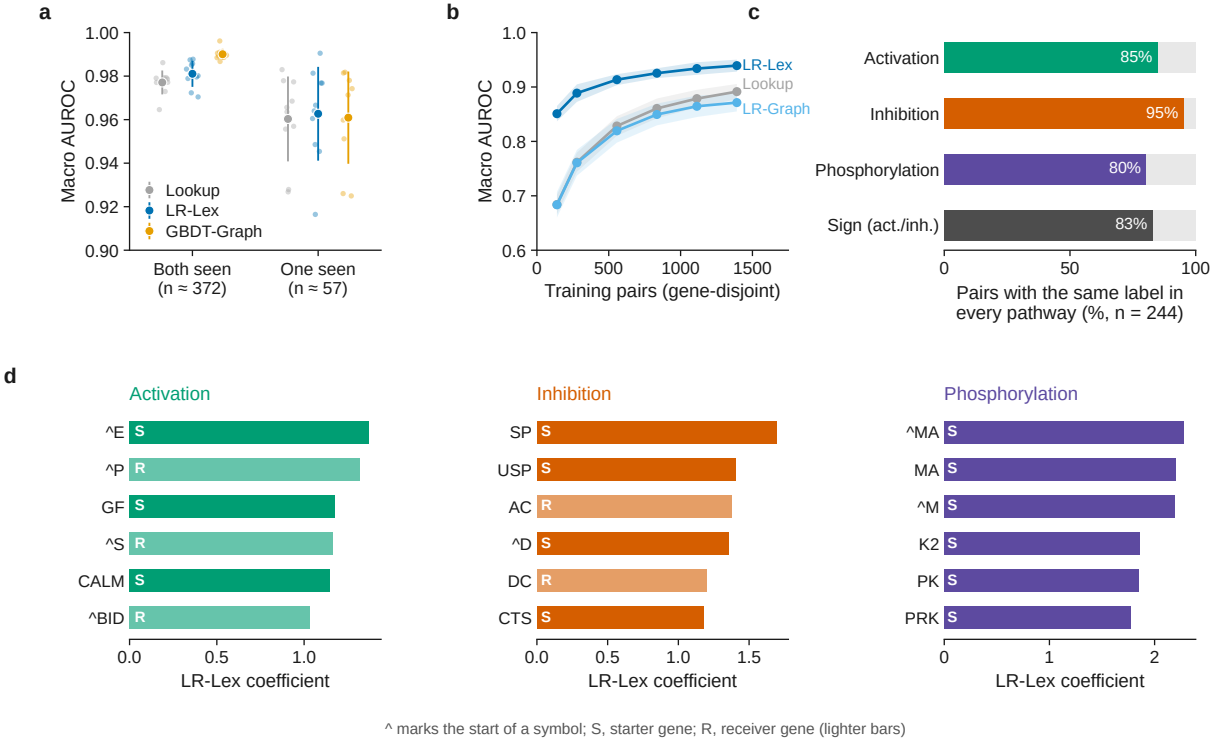
a
Macro AUROC
1.00
0.98
0.96
0.94
0.92
0.90
Lookup
LR-Lex
GBDT-Graph
Both seen
(n ≈ 372)
One seen
(n ≈ 57)
b
Macro AUROC
1.0
0.9
0.8
0.7
0.6
LR-Lex
Lookup
LR-Graph
0
500
1000
1500
Training pairs (gene-disjoint)
c
Activation
85%
Inhibition
95%
Phosphorylation
80%
Sign (act./inh.)
83%
0
50
100
Pairs with the same label in
every pathway (%, n = 244)
d
Activation
^E
S
^P
R
GF
S
^S
R
CALM
S
^BID
R
0.0
0.5
1.0
LR-Lex coefficient
Inhibition
SP
S
USP
S
AC
R
^D
S
DC
R
CTS
S
0.0
0.5
1.0
1.5
LR-Lex coefficient
Phosphorylation
^MA
S
MA
S
^M
S
K2
S
PK
S
PRK
S
0
1
2
LR-Lex coefficient
^ marks the start of a symbol; S, starter gene; R, receiver gene (lighter bars)